Linear Equations in Linear systems

Example

system of **linear equations**:

$$\begin{array}{rcrcrcr} x_1 & - & 2x_2 & - & 7x_3 & = & -1 \\ -x_1 & + & 3x_2 & + & 6x_3 & = & 0 \end{array}$$

- variables: x_1, x_2, x_3.
- coefficients:

$$\begin{array}{rcrcrcr} {\color{red}1}x_1 & - & {\color{red}2}x_2 & - & {\color{red}7}x_3 & = & -1 \\ {\color{red}-1}x_1 & + & {\color{red}3}x_2 & + & {\color{red}6}x_3 & = & 0 \end{array}$$

- constant terms:

$$\begin{array}{rcrcrcr} x_1 & - & 2x_2 & - & 7x_3 & = & {\color{red}-1} \\ -x_1 & + & 3x_2 & + & 6x_3 & = & {\color{red}0} \end{array}$$

$x_1 = -3$, $x_2 = -1$, $x_3 = 0$ is a solution to the system

$$\begin{array}{rcrcrcr} x_1 & - & 2x_2 & - & 7x_3 & = & -1 \\ -x_1 & + & 3x_2 & + & 6x_3 & = & 0 \end{array}$$

as is $x_1 = 6$, $x_2 = 0$, $x_3 = 1$. However, $x_1 = 0$, $x_2 = 0$, $x_3 = 0$ is not a solution to the system.
A **solution to the system** must be a solution to **every equation** in the system. The system above is **consistent**, meaning that the system has **at least one** solution. Here is an example of an **inconsistent** system, meaning that it has no solutions.

$$\begin{array}{rcrcrcr} x_1 & + & x_2 & + & x_3 & = & 0 \\ x_1 & + & x_2 & + & x_3 & = & -8 \end{array}$$

Why is the above system inconsistent?

Graphical Solutions

Example

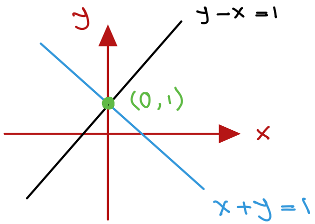
Consider the system of linear equations in two variables

$$\begin{aligned} x + y &= 1 \\ y - x &= 1 \end{aligned}$$

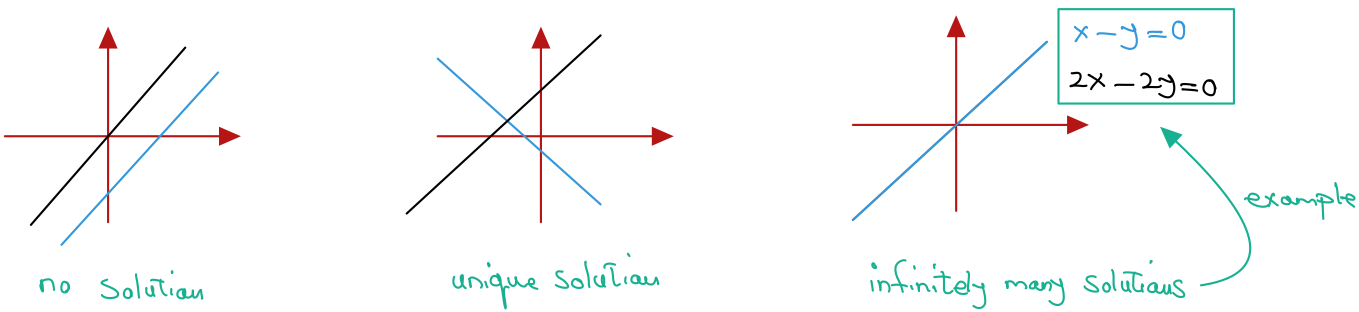
A solution to this system is a pair of real numbers (x, y) satisfying these two equations. Graphically, a solution to this system is the point of intersection of these two lines.

add the eqs: $2y = 2$ $\boxed{y=1}$

Sub $y=1$ into one of the eqs: $x+1=1$ $\boxed{x=0}$



Given a system of two equations in two variables, graphed on the xy-coordinate plane, there are three possibilities:



Example

The system of linear equations in three variables that we saw earlier

$$\begin{array}{rcrcrcr} x_1 & - & 2x_2 & - & 7x_3 & = & -1 \\ -x_1 & + & 3x_2 & + & 6x_3 & = & 0, \end{array}$$

has solutions $x_1 = -3 + 9s$, $x_2 = -1 + s$, $x_3 = s$ where s is any real number ($s \in \mathbb{R}$). s is called a parameter, and

$$x_1 = -3 + 9s, x_2 = -1 + s, x_3 = s, \text{ where } s \in \mathbb{R}$$

is called the solution set in parametric form.

Question: What is the graphical interpretation of the solution set in this example?

The intersection of Two planes is a line whose Parametric eqs. are The Solution set for The above SLEs.

Problem

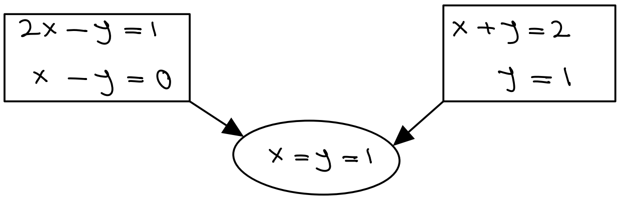
Find all solutions to a system of m linear equations in n variables, i.e., solve a system of linear equations.

Definition

Two systems of linear equations are equivalent if they have **exactly the same** solutions.

Example

The following SLEs have the same solutions and hence they are equivalent.



We solve a system of linear equations by using *Elementary Operations* to transform the system into an equivalent but simpler system from which the solution can be easily obtained.

Elementary Operations

1. Interchange two equations.
2. Multiply an equation by a nonzero number.
3. Add a multiple of one equation to a different equation.

Example

Consider the system of linear equations
$$\begin{array}{rcrcrcr} 3x_1 & - & 2x_2 & - & 7x_3 & = & -1 \\ -x_1 & + & 3x_2 & + & 6x_3 & = & 1 \\ 2x_1 & & & - & x_3 & = & 3 \end{array}$$

- Interchange first two equations

$$\begin{array}{rcrcrcr} -x_1 & + & 3x_2 & + & 6x_3 & = & 1 \\ 3x_1 & - & 2x_2 & - & 7x_3 & = & -1 \\ 2x_1 & & & - & x_3 & = & 3 \end{array}$$

- Multiply first equation by -2

$$\begin{array}{rcrcrcr} -6x_1 & + & 4x_2 & + & 14x_3 & = & 2 \\ -x_1 & + & 3x_2 & + & 6x_3 & = & 1 \\ 2x_1 & & & - & x_3 & = & 3 \end{array}$$

- Add 3 times the second equation to the first equation

$$\begin{array}{rcrcrcr} 0 & + & 7x_2 & + & 11x_3 & = & 2 \\ -x_1 & + & 3x_2 & + & 6x_3 & = & 1 \\ 2x_1 & & & - & x_3 & = & 3 \end{array}$$

The Augmented Matrix

Example

The system of linear equations

$$\begin{array}{rcrcrcr} x_1 & - & 2x_2 & - & 7x_3 & = & -1 \\ -x_1 & + & 3x_2 & + & 6x_3 & = & 0 \end{array}$$

is represented by the augmented matrix

$$\left[\begin{array}{rrr|r} 1 & -2 & -7 & -1 \\ -1 & 3 & 6 & 0 \end{array}\right]$$

Two other matrices associated with a system of linear equations are the coefficient matrix and the constant matrix.

$$\begin{bmatrix} 1 & -2 & -7 \\ -1 & 3 & 6 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Example

Consider the system of linear equations

$$\begin{array}{rcrcrcr} 3x_1 & - & 2x_2 & - & 7x_3 & = & -1 \\ -x_1 & + & 3x_2 & + & 6x_3 & = & 1 \\ 2x_1 & & & - & x_3 & = & 3 \end{array}$$

$$\left[\begin{array}{rrr|r} 3 & -2 & -7 & -1 \\ -1 & 3 & 6 & 1 \\ 2 & 0 & -1 & 3 \end{array}\right]$$

- Interchange first two equations (two rows)

  $r_1 \leftrightarrow r_2$

$$\left[\begin{array}{rrr|r} -1 & 3 & 6 & 1 \\ 3 & -2 & -7 & -1 \\ 2 & 0 & -1 & 3 \end{array}\right]$$

- Add 3 times the second equation to the first equation

  $3r_2 + r_1$

$$\left[\begin{array}{rrr|r} 0 & 7 & 11 & 2 \\ -1 & 3 & 6 & 1 \\ 2 & 0 & -1 & 3 \end{array}\right]$$

Row-Echelon Matrix:

- All rows consisting entirely of zeros are at the bottom.
- The first nonzero entry in each nonzero row is called a leading entry.
- Each leading entry is to the right of all leading entries in rows above it.

Example

$$\begin{bmatrix} 0 & \square & * & * & * & * & * & * \\ 0 & 0 & 0 & \square & * & * & * & * \\ 0 & 0 & 0 & 0 & \square & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & \square & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\square$ can be any non-zero real number.

$*$ can be any real number.

Reduced Row-Echelon Matrix:

- Row-echelon matrix.
- The leading entry in each nonzero row is 1.
- Each leading 1 is the only nonzero entry in its column.

Example

$$\begin{bmatrix} 0 & 1 & * & 0 & 0 & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$*$ can be any real number.

Example

Suppose that the following matrix is the augmented matrix of a system of linear equations.

$$\left[\begin{array}{ccccccc|c} 1 & -3 & 4 & -2 & 5 & -7 & 0 & 4 \\ 0 & 0 & 1 & 8 & 0 & 3 & -7 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

Note that the matrix is a row-echelon matrix.

- A pivot position is a location corresponding to a leading 1 and a pivot column is the column containing the pivot position.
- The basic variables are the variables that correspond to columns containing leading ones.
- The remaining variables (columns 2, 5 and 6) are called free variables.

We will use elementary row operations to transform a matrix to row-echelon or reduced row-echelon form.